

Spiral diopter: freeform lenses with enhanced multifocal behavior

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Lens design is of paramount importance in the evolving world of technology, where compactness and high optical performance are a necessity, ranging from smartphones and wearable devices to vehicles and virtual reality. Freeform design techniques allow us to transcend traditional limitations, but creating new optics remains a substantial challenge unless we consider unconventional physical phenomena. Here, we introduce a lens type based on freeform design, employing spiralization of one of its diopters that results in optical vortices. This enables multifocality, primarily serving needs in ophthalmology; however, its potential applications could broadly impact many other domains. In particular, this lens design could be crucial in miniaturizing emerging technologies while retaining their optical quality. © 2024 Optica Publishing Group under the terms of the [Optica Open Access Publishing Agreement](https://doi.org/10.1364/OPTICA.507066)

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1. INTRODUCTION

Within the domain of imaging applications, the use of multifocality and an extended depth of field (EDOF), also known as point spread function (PSF) engineering, has emerged as a novel paradigm [1–4]. These techniques offer extensive advantages and unparalleled capabilities, thus finding widespread usage, particularly in the fields of computational imaging and microscopy. Depth of field multiplexing (DOFM) has been instrumental in this regard [5–13].

For instance, in single molecule localization microscopy (SMLM), DOFM is employed to enhance the Fisher information along the optical axis [5–10]. On the other hand, in computational imaging, DOFM facilitates the capture of a depth image with a monocular system using a single exposure. This is achieved through modifications to the optical system accompanied by suitable algorithmic approaches [11–13].

Either EDOF or DOFM is then achieved by inserting an optical system capable of implementing PSF engineering, which is less constrained. Several approaches can be used, such as the use of cylindrical lenses [5], phase masks [10], deformable mirrors (DM)s [14], and spatial light modulators (SLM) [7,15].

However, the prospective usage of multifocality and EDOF extends to other diverse fields where these approaches are not compatible with the trend toward miniaturization required by emerging technologies for smartphones, wearables, automotive, and virtual reality, which cannot rely on such devices, but seek an optimization of the form factor while maintaining the optical qualities [16]. This is the case in monocular imaging, where coded aperture [11,12] or freeform lenses [1] are used. However, the

design of such systems remains quite complex to optimize. This is particularly true for the freeform approach for which the problem of parameterization remains problematic [17,18].

Achieving EDOF or multifocality can be of importance in ophthalmology, where these features ascend in importance for correcting presbyopia, especially when associated with other ocular conditions [19], as well as during the implantation of intraocular lenses after cataract extraction [20,21]. However, the design of optics in ophthalmology is constrained by the size and wearability of contact lenses and intraocular implants. Multifocality is commonly achieved through diffractive or refractive designs, primarily based on modifications to the pupil through ring patterns. While effective, they can make the eye/lens system sensitive to the iris opening, causing potential issues, especially in photopic vision [22,23]. Light sword lenses [24,25], while impressive in delivering an extended depth of field possibly independently from the aperture, may sometimes be inadequate due to their physical structure, which includes a groove. This design limitation can obstruct miniaturization efforts, restrict their use to small diameters and clean environments, and may hinder comfort in contact lens applications due to possible interference with eyelids.

In the realm of mixed reality, it is necessary for a display system to adapt to the distance of the visualized objects due to the problem of the vergence-accommodation conflict [26,27]. Common approaches aim to implement dynamic adaptation systems, which in turn lead to a highly complex device design. In this context, projection lenses with a very large depth of field may represent a passive, comparatively inexpensive, and easy-to-implement solution [28].

In this paper, we present what we believe, to the best of our knowledge, is a new concept for freeform multifocal lenses, first approached by Jones and Clutterbuck in 2003 [29] and fully formalized and realized by Galinier in 2019 [30]. This concept allows for shaping the focal points (position and depth of field) from a few simple parameters on the diopter geometry, offering the possibility of choosing multifocal behavior, extended depth of field, and/or depth encoding through the shapes of the focal spots, independent of the aperture. To the best of our knowledge producing such behavior from a simple lens is unique and represents a breakthrough within the conception of multifocal optical surfaces. First, we will present the original concept of these new lenses and their surfaces/diopters. Second, we will numerically simulate their behavior and compare these simulations with experimental results obtained from lenses we have manufactured. From these preliminary results, we will propose a physical interpretation of their behavior based on the presence of optical vortices in the focal areas.

2. DESIGN AND MODELING

From the point of view of optical engineering, astigmatism is generally considered a defect. However, astigmatic lenses have the particular property of encoding several focal lengths in a single pupil, thanks to the presence of different curvatures along their tangential and sagittal axes. The innovation behind Galinier's

lens design [30], which we discuss here, consists in exploiting the properties of astigmatic lenses while attenuating their resolution and anisotropy defects by arranging the distribution of focal power within the lens itself. Our original idea involves breaking away from symmetrical lens designs that rely on uniform focal distribution, and instead embracing a freeform design approach that is achieved by distributing the focal powers on the output diopter according to a Fermat spiral pattern, which divides the surface into two equal zones. As a result, we expect the modified diopter to provide two equivalent optical fields that converge in two different zones, interfering to produce new focal zones. As depicted in Fig. 1(a), traditional astigmatic lenses exhibit several focal regions with resolution and anisotropy defects. In contrast, our modified lens, shown in Fig. 1(b), exhibits a multifocal region through the spiralization of the lens. This approach is expected to improve image quality in the obtained focal areas, as illustrated by the associated PSFs.

To combine two quantities, c_1 and c_2 (awaited optical powers of the lens), following a Fermat spiral on a disk with radius R , we can use the relationship

$$\mathcal{C}(\Phi, \rho; c_1, c_2; N, \eta) = \frac{c_1}{2} [1 + \cos(N\Phi + \eta\hat{\rho}^2)] + \frac{c_2}{2} [1 - \cos(N\Phi + \eta\hat{\rho}^2)]. \quad (1)$$

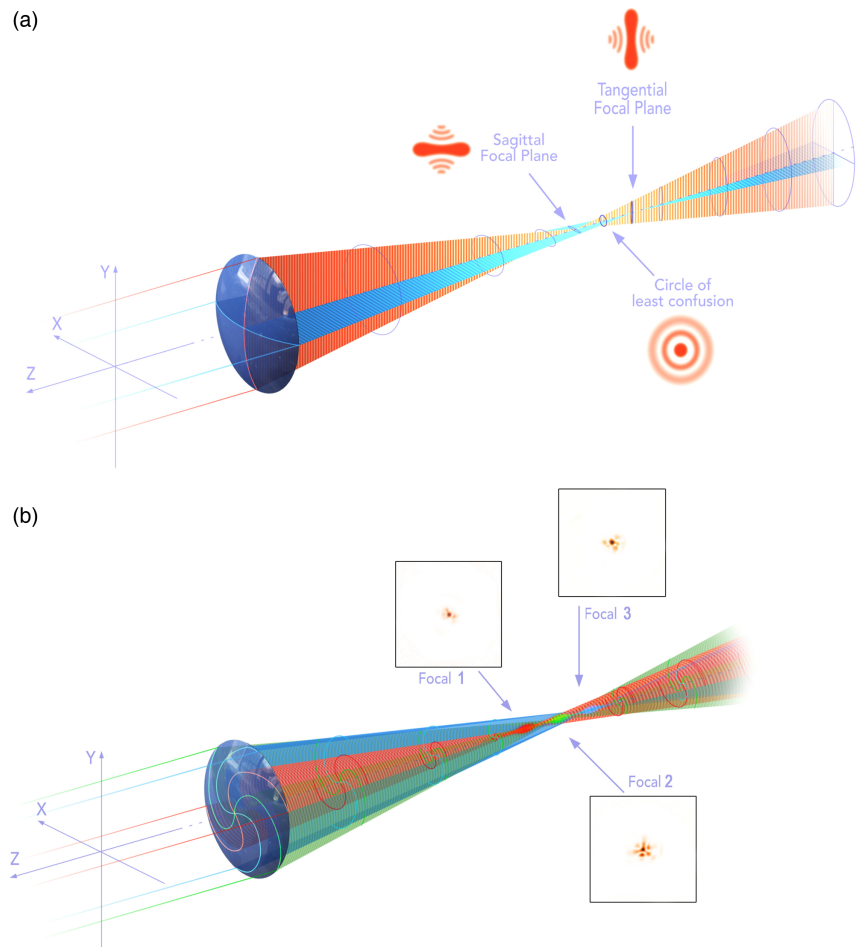


Fig. 1. Comparison between (a) a conventional astigmatic lens and (b) our spiral lens. For each lens a ray tracing representation is given as well as a representation of the PSF in each expected focal zone.

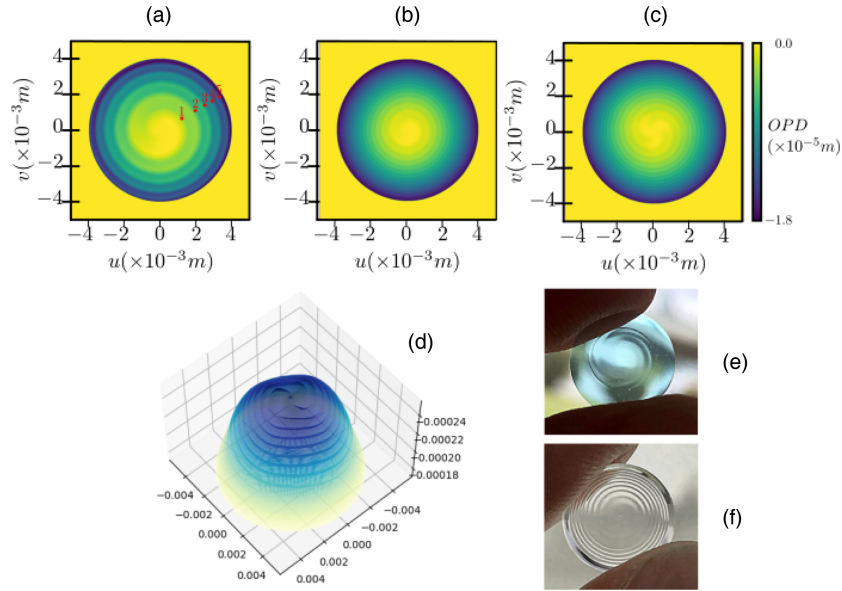


Fig. 2. Spiralized combination of focals from two astigmatic lenses. (a)–(c) Lenses OPD obtained for different couples of parameters (N, η) . (a) $N = 1, \eta = 5$. The number of turns can be evaluated by counting the grooves present on a radius (points and numbers in red). (b) $N = 1, \eta = 15$; (c) $N = 3, \eta = 5$: The resulting number of grooves is similar to (b). (d) Resulting OPD plotted path difference compared to spherical surface for $\eta = 5$ and $N = 3$. (e) and (f) Two photographs of prototypes of rigid contact lenses that we have manufactured: (e) spiralized meniscus with $N = 5$ and a logarithmic spiral, $R_1 = 7.91$ mm, $R_2 = 7.5$ mm, and $R_{\text{back}} = 7.9$ mm; the thickness is 0.5 mm; and total diameter is 10 mm. (f) Spiralized meniscus with $N = 1, \eta = 10$ and a Fermat spiral, $R_1 = 7.91$ mm, $R_2 = 7.5$ mm, $R_{\text{back}} = 7.9$ mm; the thickness is 0.5 mm and the total diameter is 10 mm.

In this equation, Φ and ρ represent the polar coordinates, N denotes the number of spiral branches, η represents the total number of turns, and $\hat{\rho}$ is the normalized radius given by $\hat{\rho} = \rho / R$ with $\hat{\rho}$ ranging from 0 to 1. The function $\mathcal{C}$ represents a distribution of the power within the pupil of the lens in diopters, giving, under the paraxial assumption [31], the lens phase function

$$\mathcal{W}_l(\rho, \phi) \equiv \exp \left[-jk \frac{\rho^2}{2} \mathcal{C}(\rho, \Phi; c_1, c_2; N, \eta) \right]. \quad (2)$$

To physically realize the optical power distribution $\mathcal{C}$ in a lens, it thus becomes necessary to manipulate the curvatures of at least one of the diopters. This manipulation leads to the creation of a freeform surface that can generate the desired multifocal behavior described in [30]. Figure 2 illustrates the combination by spiralization and the resulting phase addition. Figures 2(a)–2(c) give the distribution maps of the combined quantities according to different pairs of parameters (N, η) while Fig. 2(d) represents the sag obtained after the spiralization of two radii of curvature $7.91 \cdot 10^{-3}$ m and $7.5 \cdot 10^{-3}$ m and parameters $N = 3$ and $\eta = 5$. In the lower right corner, two rigid contact lenses that have undergone spiralization are displayed. These contact lenses were produced using a Optoform 40 FTS 5000 digital lathe, powered by a bespoke driving algorithm that we developed. Examples of the obtained lenses are shown in Figs. 2(e) and 2(f). Subsequently, we conducted a characterization of these lenses to demonstrate that they exhibit the anticipated behavior.

3. SIMULATION AND EXPERIMENTAL MEASUREMENTS

We have specifically designed a 10 mm diameter contact lens to facilitate its comprehensive evaluation through simulation techniques and on a focometry bench. The lens is engineered in PMMA

(index $n = 1.49$) to combine $R_1 = 8.18$ mm and $R_2 = 7.8$ mm curvature radii on its outer dioptr as well as $R_{\text{back}} = 8.0$ mm on its inner dioptr. It was spiralized with $N = 1$ and $\eta = 15$ using a 7.2 mm diameter aperture (the effective optical zone). To assess its performance, the lens is evaluated under a 532 nm wavelength illumination with two modalities. First, simulation-based evaluation is conducted by propagating the exit pupil using the algorithm proposed by Schmidt [32]. Additionally, experimental evaluation is undertaken using a designed focometry bench.

From the phase function given in the previous section in Eq. (2), it is possible to compute its generalized pupil function. Considering a unit-amplitude, normally incident, quasi-monochromatic plane wave, it is then possible to perform a numerical propagation of the optical field [31]. Several approaches are possible to evaluate the optical field after the lens. As we expect a multifocal behavior, the focal length is not determined, and we need to observe the field all along the optical axis. Consequently, the Fraunhofer diffraction of the generalized pupil method is clearly not suitable. The computation of the propagation of an optical field over large distances involves sampling problems that concern the volumes to be simulated, the desired resolution as well as the sampling problems on the phase distribution of the propagator. Moreover, in our case, the sampling problem of the entrance pupil becomes particularly critical since it obeys a function of type $\cos(a + b\hat{\rho}^2)$. An elegant solution would be to perform an n steps propagation [32,33] to obtain the scalar optical field on each plane along the lens axis. The field $U(\mathbf{r}_n)$ at the observation plane located at the distance z_n can be expressed by Eq. (3):

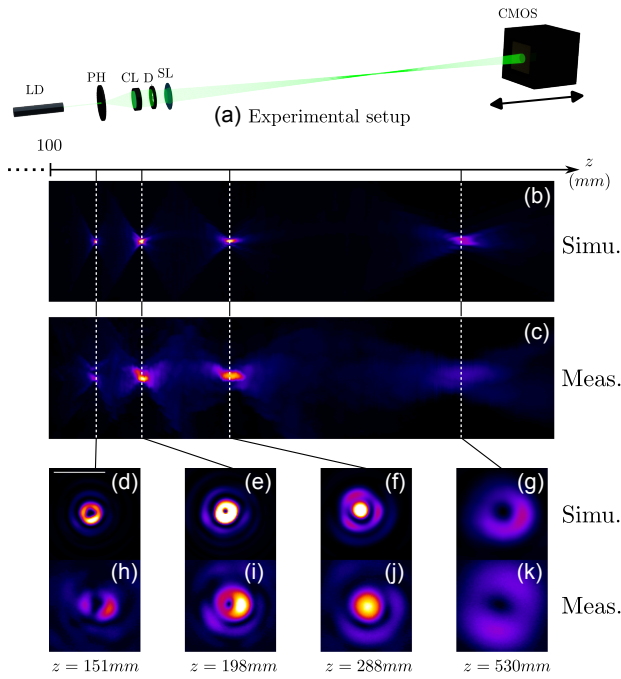


Fig. 3. Simulations and experimental measurements. (a) Experimental setup for PSF measurement along the z axis: LD, laser diode (532 nm) source; PH, 40 μm pinhole; CL, $f = 300$ mm collimating lens; D, aperture diaphragm; SL, studied spiralized lens; and CMOS, CMOS sensor traveling along the z axis thanks to a worm screw driven by a stepper motor. Representation of PSFs obtained by the simulations shown in (b), (d)–(g) and by experimental measurements shown in (c), (h)–(k). (b) and (c) The maximum intensity profiles in $x - z$ and (d)–(k) the $x - y$ slices obtained for the detected focus areas for $z = 151$ mm, $z = 198$ mm, $z = 288$ mm, and $z = 530$ mm; scale bar in (d) is 100 μm .

$$U(\mathbf{r}_n) = \mathcal{Q} \left[\frac{m_{n-1} - 1}{m_{n-1} \Delta z_{n-1}}, \mathbf{r}_n \right] \prod_{i=1}^{n-1} \left\{ \mathcal{A}[\mathbf{r}_{i+1}] \mathcal{F}^{-1} \left[\mathbf{f}_i, \frac{\mathbf{r}_{i+1}}{m_i} \right] \mathcal{Q}_2 \left[-\frac{\Delta z_i}{m_i}, \mathbf{f}_i \right] \mathcal{F} \left[\mathbf{r}_i, \mathbf{f}_i \right] \frac{1}{m_i} \right\} \left\{ \mathcal{Q} \left[\frac{1 - m_1}{\Delta z_1}, \mathbf{r}_1 \right] U(\mathbf{r}_1) \right\}, \quad (3)$$

where the vectors $\mathbf{r}_i$ and $\mathbf{f}_i$ point, respectively, the Cartesian coordinates of $M_i(x_i, y_i)$, and the spatial frequencies (f_{x_i}, f_{y_i}) in the i th plane while Δz_i and m_i represent, respectively, the distance and the scaling factor (due to numerical Fourier transform process) from the i th to $(i + 1)$ th plane. $\mathcal{Q}$ and $\mathcal{F}$ correspond to the operators introduced by Goodman [31] as, respectively, the quadratic phase exponential multiplication and the Fourier transformation. $\mathcal{A}$ corresponds to the filtering window (e.g., super-Gaussian or Tukey) applied to the plane before the Fourier transform. We have implemented the calculation of this expression in Python language according to an iterative process to obtain the optical field beyond the lens that we want to simulate. The calculation is made from the parameters of the lens (curvature, index, and spiralization parameters). The results of the simulations are presented in Figs. 3(b), and 3(d)–3(g).

Our measurement setup is designed to be relatively simple, as shown in Fig. 3(a). It employs a 532 nm laser diode as the light source, which is directed through a 40 μm pinhole. The pinhole size was chosen to be smaller than half of the expected resolution for the aperture and focal length under study. The collimated beam produced by a collimation lens (Thorlabs AC508-300-AB, $f = 300$, 0 mm) traverses a diaphragm-controlling system aperture and illuminates the entrance diopter of our contact lens

(i.e., the spiralized lens). We use a CMOS camera (IDS u3-3890cp-m-gl) with square pixels of 1.85 μm mounted on a motorized stage that moves along a worm gear driven by a stepper motor. The entire setup is interfaced and controlled using LabVIEW software, which allows us to capture a series of images of the optical field intensity downstream of the lens with a precision of 1/10th of a millimeter for each 1 mm step. The outcomes yielded from the experiments are showcased and juxtaposed with the simulation, as delineated in Figs. 3(c), and 3(h)–3(k).

4. RESULTS AND DISCUSSION

Figure 3 presents the intensity distributions of the optical field downstream from the lens, obtained through both the simulation [Figs. 3(b) and 3(d)–3(g)] and the experimental characterization [Figs. 3(c) and 3(h)–3(k)]. Figures 3(d)–(k) depict the $x - y$ slices of the 3D optical field intensity for both the simulated in Figs. 3(d)–(g) and the experimental measurement in Figs. 3(h)–(k).

The maximum intensity projections along the y axis, displayed in Figs. 3(b) and 3(c), allow the identification of the focal zones emerging from the lens in both the simulation and the measurement. In the simulation in Fig. 3(b), we observe four distinct focal regions at distances of 151, 198, 288, and 530 mm. The presence of more than two focal regions can be attributed to the continuous nature of the $\mathcal{C}$ function, which allows for intermediate focal zones.

These four focal regions are also observed in the experimental measurements; however, their PSF shape does not perfectly match the simulated results. This discrepancy can be attributed to two main factors. First, the limitations of the sensor's dynamics prevent it from precisely reproducing the intensity distributions obtained from the simulation. Second, the manufacturing process introduces tolerances that inevitably impact the quality of the results.

Furthermore, maintaining the alignment of the setup over such a large distance is challenging, as we observed a minor angular deviation from the optical axis, necessitating digital realignment of the recorded images. Nevertheless, considering the angles involved, it is not feasible to adjust the measured intensities accordingly.

From these results, two important conclusions can be drawn. First, the fabricated lens exhibits the expected behavior in terms of producing multiple focal zones. Second, the experimental measurements validate the simulation, provided we acknowledge a certain level of deviation that can be attributed to manufacturing defects. After these initial results, we would like to characterize more specifically the multifocal behavior of the spiral lens. More specifically, we aim to use simulation to compare its performance with that of a conventional trifocal lens that has been specifically parameterized to match the foci of the spiral lens under study and optimized to limit aberrations. Additionally, we will investigate the influence of the parameter η on the optical characteristics of the spiral lens.

Figure 4 presents a comparison between the performance of a multifocal spiral lens and a conventional trifocal lens, with a focus on their optical field intensity, modulation transfer function (MTF) properties, and the effect of η parameters on the position

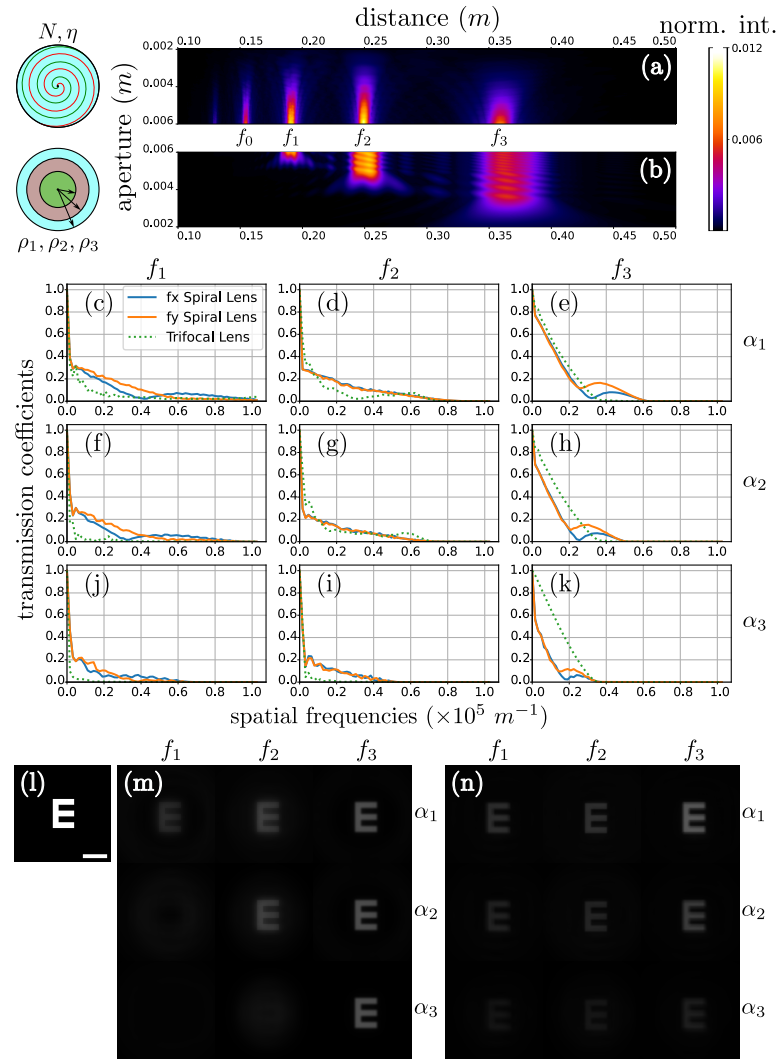


Fig. 4. Simulation: Comparison of the multifocal spiral lens ($N=1$, $\eta=16$, 5 , $R_1=8.28$ mm, and $R_2=7.86$) with a classic trifocal lens ($R_1=8.28$ mm for $0 < \rho \leq 3.47$ mm, $R_2=8.11$ mm for $3.47 < \rho \leq 4.9$ mm, and $R_1=8.28$ mm for $4.9 < \rho \leq 6$ mm). (a) and (b) Maximum intensity projection of the optical field along the z axis following its passage through a trifocal lens where the horizontal axis represents the propagation distance (focal length), and the vertical axis shows the changes in the aperture (diaphragm in front of the lens) for (a) a lens with a spiralized diopter and (b) a classical trifocal lens. (c)–(k) Modulation transfer function (MTF) curves computed for both lenses at the optimal focal zones (c), (f), (i) f_1 ; (d), (g), (j) f_2 ; and (e), (h), (k) f_3 , for three apertures values α : (c)–(e) $\alpha_1 = 6$ mm, (f)–(h) $\alpha_2 = 4.9$ mm, and (i)–(k) $\alpha_3 = 3.47$ mm. The MTF is depicted as green dashed lines for the trifocal lens and as solid blue lines along the f_x axis or orange lines along the f_y axis for the spiral lens. (l)–(n) Simulated images of (l) obtained for (m) the trifocal lens and (n) the spiral lens computed for the different optimal focal zones (f_i) and apertures (α_j). The calibration bar in (l) is 5 arcmin.

of focal points along the optical axis of the spiral lens. The MTFs are obtained by simply computing the modulus of the PSF Fourier transform. Figures 4(a) and 4(b) show the projection of the maximum optical field intensity downstream of each lens along the z axis, as a function of propagation distance and aperture size. Three focal zones, denoted as f_1 , f_2 , and f_3 , are observed in both cases. The spiral lens exhibits an additional focusing zone (f_0) and a more interesting multifocal behavior even at lower aperture sizes, with focal zones f_1 and f_2 maintaining satisfactory intensity. This indicates a wider range of trifocal behavior compared to the concentric ring trifocal lens. The MTF curves in Figs. 4(c)–4(k) further support these findings. For the spiral lens, MTFs at the optimal focal zones of f_1 , f_2 , and f_3 generally show better performance compared to the trifocal lens, particularly in Figs. 4(c), 4(f), and 4(i). The spiral lens exhibits comparable MTFs to the trifocal lens in Figs. 4(d), 4(e), 4(g), and 4(h), while the trifocal lens

displays a more favorable outcome in Fig. 4(k), albeit at a cutoff frequency above $0.2 \times 10^{-5} \text{ m}^{-1}$. However, it's important to note that given the aperture considered ($\alpha_3 = 3.47$ mm), the trifocal lens can no longer be considered as such, since only the central zone corresponding to a single focal length is used. The comparison to the spiral lens, which is still multifocal despite its small aperture, is therefore unfair. The calculated images shown in Figs. 4(l)–4(n) provide excellent support for the conclusions drawn from the MTF analysis, demonstrating the interesting capabilities of our spiral lens.

These results provide valuable insights into the performance of the spiral lens compared to the classic trifocal lens. The comparable MTFs in selected scenarios indicate that the spiral lens can achieve a similar performance to the trifocal lens, while outperforming it in other situations. An intriguing observation is the presence of an additional focal point (f_0) for the spiral lens, even though

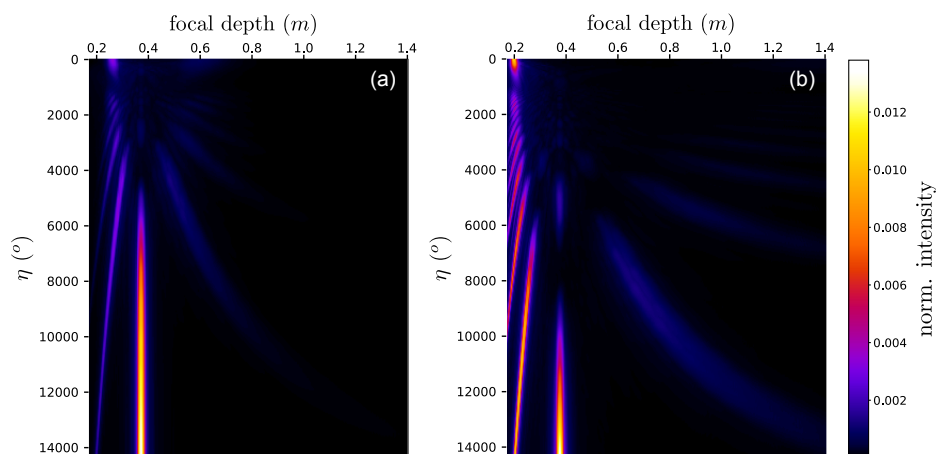


Fig. 5. Changing focal positions based on η (in degrees) for two given spiral lenses. (a) and (b) Computed maximum intensity projections of the optical field intensity on the z -axis depending on the propagation depth (horizontal axis) and the spiralization parameter η (vertical axis) for two lenses combining curvatures R_1 and R_2 , for which the relative difference is (a) $\Delta R/1, 2 \approx 0.28$ mm and (b) $\Delta R/1, 2 \approx 0.56$ mm.

it has been designed from only two curvatures. To delve deeper into this phenomenon, we have conducted supplementary simulations. Longitudinal maximum intensity projections presented in Figs. 5(a) and 5(b) are obtained by continuously varying the η parameter of the spiralization for two pairs of associated curvatures. An original behavior is observed, with focal points shifting along the z axis depending on the number of turns in the spiral. The larger the $\Delta R_{1,2}$, the faster the shift of focal points. These projections highlight the fact that, depending on η and $\Delta R_{1,2}$, we can observe several focal points ranging from one to three or four. This demonstrates that the position and number of focal points can be achieved simply by manipulating these parameters. However, it should be noted that the focal points vary in terms of their intensity; therefore, further optimization is required.

Finally, to gain a deeper understanding of the specific behavior exhibited by these spiral lenses, one can carefully examine the experimental results in Figs. 3(d), 3(e), and 3(g) as well as the simulations Figs. 3(i) and 3(k). Notably, dark spots in the center of the PSF can be observed. Such dark spots are indicative of the presence of optical vortices [34–36]. This can be explained by reconsidering the way spiral lenses are constituted. The wavefront downstream of the lens is determined by the thickness of the lens at each point in its aperture, which can be associated with a phase function. For a classical lens, the phase function is typically approximated by a quadratic phase function; this, however, changes upon introducing an angular dependency to the phase (i.e., a dependency on Φ), as seen in light sword lenses. These lenses blend a quadratic phase function with an angular dependency, introducing a spiral phase plate function [37]. Distinctively, our spiral lenses also introduce a unique radial dependency, as shown in Eqs. (1) and (2). This function combination has previously been studied with structures like fractal zone plates, Dammann zone plates, and lens-phase masks, exhibiting multifocal behavior as well as high-quality optical vortices [38,39,40,41]. Based on this, it is rational to assume that altering the η parameter enables control of the topological charge in the focal beam's phase and fine-tuning of the multifocal behavior. Here, by integrating these features onto the diopter of a “classical” lens, we have created a compact and wearable optical element that produces such vortices with potential applications in wearable optics and ultracompact embedded imaging systems.

5. CONCLUSION

In conclusion, our study has introduced what we believe, to the best of our knowledge, is the innovative concept of a spiralized freeform multifocal lenses, offering the advantages of independent focus shaping and an extended range of focalization. By incorporating the spiral pattern into the lens diopter, we have successfully engineered a compact and transportable optical element endowed with the intriguing capability to generate optical vortices. This groundbreaking achievement opens new avenues in the realms of wearable optics and ultracompact embedded imaging systems. The spiral lenses presented show advantages over conventional trifocal lenses at larger apertures for most focal points, while preserving the multifocal behavior even at smaller apertures, a feature not exhibited by the trifocal lens. These findings underscore the potential of Galinier's freeform multifocal lenses to provide wider depth perception while reducing the reliance on larger apertures in real-world scenarios.

To further advance these lenses, future research should focus on optimizing the design parameters, such as the shape and distribution of the spiral, to enhance performance across various apertures and focal zones. Additionally, investigating the behavior of optical vortices and their response to different aberrations could expand the range of applications and further enhance overall performance. In summary, spiralized multifocal lenses represent a significant advancement in multifocal optical surfaces, combining the advantages of optical vortices with a compact and wearable design, and hold great promise for further development and application in the field.

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Data availability. The data supporting the conclusions of this article will be available upon reasonable requests.

REFERENCES

1. E. R. Dowski and W. T. Cathey, “Extended depth of field through wavefront coding,” *Appl. Opt.* **34**, 1859–1866 (1995).
2. W. T. Cathey and E. R. Dowski, “New paradigm for imaging systems,” *Appl. Opt.* **41**, 6080–6092 (2002).
3. J. N. Mait, G. W. Euliss, and R. A. Athale, “Computational imaging,” *Adv. Opt. Photonics* **10**, 409 (2018).

4. U. Akpınar, E. Sahin, M. Meem, *et al.*, "Learning wavefront coding for extended depth of field imaging," *IEEE Trans. Image Process.* **30**, 3307–3320 (2021).
5. B. Huang, W. Wang, M. Bates, *et al.*, "Three-dimensional super-resolution imaging by stochastic optical reconstruction microscopy," *Science* **319**, 810–813 (2008).
6. M. F. Juette, T. J. Gould, M. D. Lessard, *et al.*, "Three-dimensional sub-100 nm resolution fluorescence microscopy of thick samples," *Nat. Methods* **5**, 527–529 (2008).
7. S. R. P. Pavani, M. A. Thompson, J. S. Biteen, *et al.*, "Three-dimensional, single-molecule fluorescence imaging beyond the diffraction limit by using a double-helix point spread function," *Proc. Natl. Acad. Sci. USA* **106**, 2995–2999 (2009).
8. C. Maurer, S. Khan, S. Fassel, *et al.*, "Depth of field multiplexing in microscopy," *Opt. Express* **18**, 3023–3034 (2010).
9. H. Kirshner, F. Aguet, D. Sage, *et al.*, "3-D PSF fitting for fluorescence microscopy: implementation and localization application," *J. Microsc.* **249**, 13–25 (2012).
10. L. von Diezmann, Y. Shechtman, and W. E. Moerner, "Three-dimensional localization of single molecules for super-resolution imaging and single-particle tracking," *Chem. Rev.* **117**, 7244–7275 (2017).
11. A. Levin, R. Fergus, F. Durand, *et al.*, "Image and depth from a conventional camera with a coded aperture," *ACM Trans. Graph.* **26** (2007).
12. P. Trounev, F. Champagnat, G. L. Besnerais, *et al.*, "Passive depth estimation using chromatic aberration and a depth from defocus approach," *Appl. Opt.* **52**, 7152–7164 (2013).
13. Q. Guo, Z. Shi, Y.-W. Huang, *et al.*, "Compact single-shot metalens depth sensors inspired by eyes of jumping spiders," *Proc. Natl. Acad. Sci. USA* **116**, 22959–22965 (2019).
14. A. Aristov, B. Lelandais, E. Rensen, *et al.*, "ZOLA-3D allows flexible 3D localization microscopy over an adjustable axial range," *Nat. Commun.* **9**, 2409 (2018).
15. S. Jia, J. C. Vaughan, and X. Zhuang, "Isotropic three-dimensional super-resolution imaging with a self-bending point spread function," *Nat. Photonics* **8**, 302–306 (2014).
16. W. T. Chen and F. Capasso, "Will flat optics appear in everyday life anytime soon?" *Appl. Phys. Lett.* **118**, 100503 (2021).
17. K. P. Thompson and J. P. Rolland, "Freeform optical surfaces: a revolution in imaging optical design," *Opt. Photonics News* **23**(6), 30–35 (2012).
18. F. Fang, X. Zhang, A. Weckenmann, *et al.*, "Manufacturing and measurement of freeform optics," *CIRP Ann.* **62**, 823–846 (2013).
19. R. Pérez-Prados, D. P. Piñero, R. J. Pérez-Cambrodí, *et al.*, "Soft multifocal simultaneous image contact lenses: a review," *Clin. Exp. Optom.* **100**, 107–127 (2017).
20. J. L. Alio, A. B. Plaza-Puche, R. Fernández-Buenaga, *et al.*, "Multifocal intraocular lenses: an overview," *Surv. Ophthalmol.* **62**, 611–634 (2017).
21. R. Rampat and D. Gatinel, "Multifocal and extended depth-of-focus intraocular lenses in 2020," *Ophthalmology* **128**, e164–e185 (2021).
22. J. P. Jorge and L. Alió, eds., *Multifocal Intraocular Lenses, Essentials in Ophthalmology* (Springer International Publishing, 2019).
23. E. Nov, A. Rubowitz, N. Dar, *et al.*, "Visual performance of a novel optical design of a new multifocal intraocular lens," *J. Refract. Surg.* **38**, 150–157 (2022).
24. K. Petelczyc, S. Bará, A. C. López, *et al.*, "Contrast transfer characteristics of the light sword optical element designed for presbyopia compensations," *J. Eur. Opt. Soc.: Rapid Publications* **6**, 11053 (2011).
25. A. Mira-Agudelo, W. Torres-Sepúlveda, J. F. Barrera, *et al.*, "Compensation of presbyopia with the light sword lens," *Invest. Ophthalmol. Vis. Sci.* **57**, 6870 (2016).
26. J. P. Wann, S. Rushton, and M. Mon-Williams, "Natural problems for stereoscopic depth perception in virtual environments," *Vis. Res.* **35**, 2731–2736 (1995).
27. L. Marran and C. Schor, "Multiaccommodative stimuli in VR systems: problems & solutions," *Hum. Factors* **39**, 382–388 (1997).
28. Y. Zhou, J. Zhang, and F. Fang, "Vergence-accommodation conflict in optical see-through display: Review and prospect," *Results Opt.* **5**, 100160 (2021).
29. L. Marran and C. Clutterbuck, "Multifocal ophthalmic lenses," PCT patent application 10/027,602 (20 December 2001).
30. L. Galinier, "Spiral dioptré with meridians of different optical power," U.S. patent US20220244569A1 (4 August 2022).
31. J. W. Goodman, *Introduction to Fourier Optics* (Roberts & Company, 2005).
32. J. Schmidt, *Numerical Simulation of Optical Wave Propagation with Examples in MATLAB* (SPIE, 2010).
33. S. Coy, "Choosing mesh spacings and mesh dimensions for wave optics simulation," *Proc. SPIE* **5894**, 589405 (2005).
34. A. White, C. Smith, N. Heckenberg, *et al.*, "Interferometric measurements of phase singularities in the output of a visible laser," *J. Mod. Opt.* **38**, 2531–2541 (1991).
35. M. S. Soskin, V. N. Gorshkov, M. V. Vasnetsov, *et al.*, "Topological charge and angular momentum of light beams carrying optical vortices," *Phys. Rev. A* **56**, 4064–4075 (1997).
36. A. M. Yao and M. J. Padgett, "Orbital angular momentum: origins, behavior and applications," *Adv. Opt. Photonics* **3**, 161 (2011).
37. A. Kolodziejczyk, S. Bará, Z. Jaroszewicz, *et al.*, "The light sword optical element—a new diffraction structure with extended depth of focus," *J. Mod. Opt.* **37**, 1283–1286 (1990).
38. S. H. Tao, X.-C. Yuan, J. Lin, *et al.*, "Sequence of focused optical vortices generated by a spiral fractal zone plate," *Appl. Phys. Lett.* **89**, 031105 (2006).
39. W. D. Furlan, F. Giménez, A. Calatayud, *et al.*, "Devil's vortex-lenses," *Opt. Express* **17**, 21891–21896 (2009).
40. J. Yu, C. Zhou, W. Jia, *et al.*, "Generation of dipole vortex array using spiral Damann zone plates," *Appl. Opt.* **51**, 6799–6804 (2012).
41. G. Rumi, D. Actis, D. Amaya, *et al.*, "Analytical description of optical vortices generated by discretized vortex-producing lenses," *Phys. Rev. A* **97**, 063813 (2018).